Answers and talk notes

Identify Equivalent Algebraic Expressions

What your student is learning

Tonight we TEST whether two expressions are equivalent — by putting the same number into both and comparing.

Answers

Ask your student to explain before you look.

1. **A.** 13. 3x + 1 = 3 × 4 + 1 = 12 + 1 = 13 when x = 4.
2. **C.** 16. 2(x + 5) = 2(3 + 5) = 2 × 8 = 16 when x = 3.
3. **B.** They match at x = 2, so test another number. They match at x = 2, so test another number. One match is not enough to call them equivalent.
4. **D.** No, they are 14 and 30. No, they are 14 and 30. When x = 5, 2x + 4 = 14 and 6x = 30.
5. **A.** 18 and 14. 3(n + 2) = 3 × 6 = 18 and 3n + 2 = 12 + 2 = 14 when n = 4. The values are 18 and 14.
6. **C.** 2m + 12. 2(m + 6) = 2m + 12. The 2 multiplies both m and 6, so 2m + 12 is equivalent.

Talk about it

When the two values are different, what does that tell you? If they match once, are you finished?

Home mission

Pretend a sandwich costs y dollars and a drink costs $7. Pick y = 5. Ask your student to find 2(5 + 7) and 2 × 5 + 14. Do they match? Why can one number not be enough proof?

When they are stuck, ask

- What number will you put into both expressions?
- Did the parentheses change what happens first?
- One test disagreed — is that enough to decide?

Try not to say

- If it works for one number, they are equivalent.
- 3x + 2x is 5x squared.
- Test with x = 1, it is the easiest.